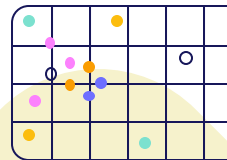
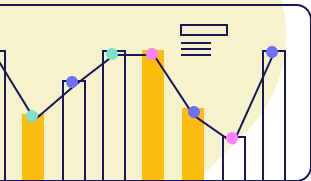


REM2 Research Methods

Lecture 5: Hedonic Price Modelling II

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What will we learn in this lecture?

01.

Assumptions behind HPM (or OLS)

Heteroskedasticity, autocorrelation, multicollinearity, ...

02.

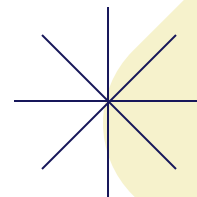
Standard diagnostic tests and checks in HPM

White test, RESET test, VIF, Scatter plots, Histograms, ...

03.

Exercises

How to use STATA or SPSS to perform the tests



OLS Assumptions – definitions, consequences and mitigations

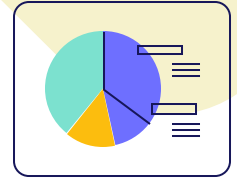
Assumptions	Consequences of violations	Mitigations
Linearity	Misspecification errors	Variable transformation (log, dummy variables, higher order terms, ...)
Homoscedasticity (Constant Variance of Errors)	Unreliable test results (t test and F test)	Estimating separate models for each sub-sample, variable transformation, interaction terms, ...
No Perfect Multicollinearity	No unique solutions. High multicollinearity causes biased coefficient estimates and inflated standard errors.	Removal of variables.
Independence (No Autocorrelation)	Biased coefficient estimates Misleading test results	Usually only a concern when using time series data. First differencing is routinely used to address this issue.
$\varepsilon \sim N(0, \sigma^2)$: The error term is normally distributed with a zero mean	Unreliable test results (t test and F test)	This assumption is less critical when the sample size is large due to the Central Limit Theorem, and the model contains an intercept term.

OLS Assumptions and Standard Diagnostic Tests

Assumptions	Standard Tests	STATA commands
Linearity	Scatter plots or residual plots, RESET test	twoway scatter ovtest
Homoscedasticity (Constant Variance of Errors)	Residual plots (residuals vs. fitted values), White Test , ...	twoway scatter estat hettest
No Perfect Multicollinearity	No need to test – the software will detect it automatically VIFs for multicollinearity (less than 10 is good)	vif
Independence (No Autocorrelation)	Durbin-Watson test and Lagrange Multiplier test	estat dwatson estat bgodfrey, lags(p)
$\varepsilon \sim N(0, a)$: The error term is normally distributed with a zero mean	Histograms, Q-Q plots, or tests like the Shapiro-Wilk test	histogram qnorm sktest

Exercises

1. Open “Lectures 4&5 Data.xlsx” in a software of your choice. Estimate the following regression model by using the rental market data only.
2. Generate the predicted values and residuals from the regression model.
3. Generate scatter plots between the residual series and the predicted values. Does the pattern appear to be random?
4. Generate a histogram of the residuals. Does it seem to be normally distributed? Perform a Q-Q test to double check the results.
5. Perform a RESET test to check if any higher order terms or interaction terms of the independent variables are missing from the regression model.
6. Request VIF statistics to check the level of multicollinearity. Is there any concern?
7. Perform a White test to check the homoskedasticity assumption. Request the Robust Standard Errors if the model does not pass the test.



Further Questions?